Zhang 2019 Follicular Lymphoma — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 4 samples (2 patients) intermix on one continuous connected manifold at Harmony θ=10 (θ=2 = per-patient archipelago); 0 detached islands; iLISI 1.04→2.26 (max 4).

Gate 2 — module geography → GO (non-branching)

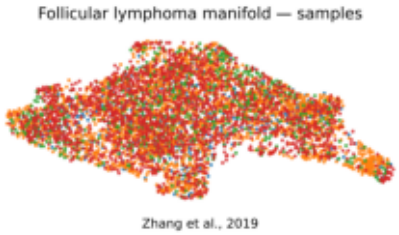
Gast/pSom weakly overlap at one pole (r=−0.16); Crest opposes both (r=−0.54 / −0.57) as an inverse terminal gradient rising along pseudotime (ρ=+0.33). No competing arms; no Crest inversion.

Gate 3 — hub cross-repression → NA

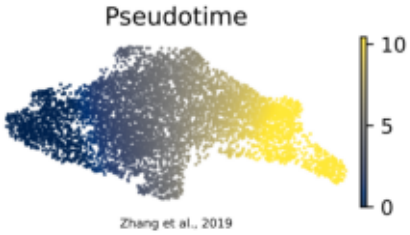
Non-branching manifold — single developmental axis, no bifurcation and no convergence hub, so the branching-only cross-repression test does not apply.

STEP FIGURES

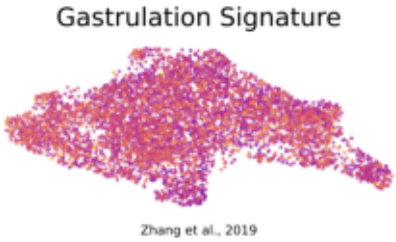
Manifold — samples



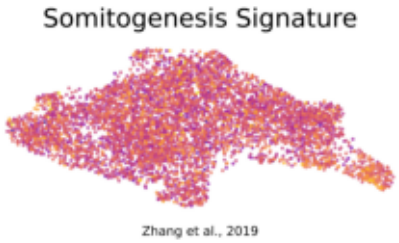
Pseudotime



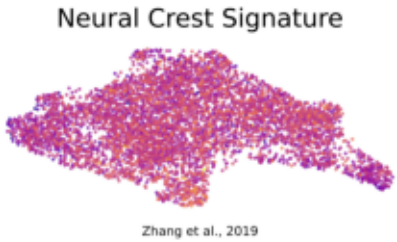
Gastrulation



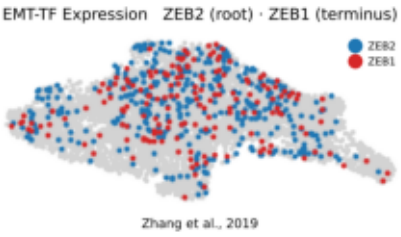
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma (Zhang et al., 2019), 10x — 4 tumour samples / 2 patients; 5,645 malignant B cells analysed.

Verdict.
The embryonic pattern reproduces in its non-branching form: a single continuous gastrulation/somitogenesis → neural-crest axis, rooted at the Gast/pSom-high pole, with EMT transcription factors ZEB2 (early) → ZEB1 (late). No bifurcation, so no hub / Gate 3 NA. Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Sample threshold	all 4 kept (min 680 >> floor; no gap cut — 2× cliff is the 2-large/2-small split)	**Integration**	Harmony θ=10 on sample, PCA(50), n_pcs=15, spread=1.3 (θ from embedding + iLISI sweep)
Island removal	radius graph 6× median-NN on UMAP; 0 cells removed (one connected component)	**Geometry**	non-branching scFates curve (Nodes=20, λ=0.3, μ=100); PPT tree tested = folds, no arm split
Root	curve tip 0 = Gast/pSom-high, Crest-low (origin; Crest late)	**EMT-TFs**	ZEB2 (early/root) ↔ ZEB1 (late/terminus), detection-gated from a 10-factor panel